Name _____ Period _____ Date _____

4-5 Lesson Quiz: Solving Rational Equations

short form

Question numbers match the Savvas Lesson Quiz. Questions 1 and 5 are not on this form. Show your work for every question: what you multiplied by, each step, and your check.

2. Aaron can paint a door in 6 hours, and Anna can do the same job in 3 hours. Working together, how many hours will it take them to paint a door?

Equation: _____

Answer: $h =$ _____ hours

Work:

3. Solve the equation $\dfrac{3}{x-1} = \dfrac{4}{2x+3}$.

Check in the original equation: _____

Solution: $x =$ _____

Work:

4. Solve the equation $\dfrac{-3}{x+3} = \dfrac{x}{x+3} - \dfrac{x}{5}$. Identify any extraneous solutions.

Solution: _____

Extraneous solution: _____

Why is it extraneous?

_____

Work:

Scoring: Question 2 and Question 4 are 10 points each. Question 3 is a bonus of up to 5 points. Total out of 20.

4-5 Lesson Quiz, short form: answer key

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2. $h = 2$ **hours.** Aaron paints $\frac{1}{6}$ of the door per hour and Anna $\frac{1}{3}$, so $\frac{1}{6}+\frac{1}{3}=\frac{1}{h}$. Multiply by $6h$: $h+2h=6$, so $3h=6$ and $h=2$. Check: $\frac{1}{6}+\frac{1}{3}=\frac{1}{2}$ door per hour, so 2 hours ✓; 2 hours is less than either time alone. Savvas quiz item 2, DOK 2, skill A87; IXL CHP (the rate equation). Taught by Example 2 (kept; Arthur and Cheyenne), rehearsed by Try It 2 and the packet's Practice 1 (TE Practice 26, Kenji 2 h and Oscar 3 h together: the same structure). **10 points:** 1 for each rate ($\frac{1}{6}$ and $\frac{1}{3}$, or equivalent), 3 for the equation $\frac{1}{6}+\frac{1}{3}=\frac{1}{h}$, 2 for clearing the fractions ($h+2h=6$), 3 for $h=2$ hours. An answer of 4.5 hours (the average of 6 and 3) or 9 hours (the sum) earns at most the 2 rate points.

3. $x=-6.5$ $\left(-\frac{13}{2}\right)$. Domain: $x\neq 1$ and $x\neq -\frac{3}{2}$. Multiply both sides by $(x-1)(2x+3)$: $3(2x+3)=4(x-1)$, so $6x+9=4x-4$, $2x=-13$, $x=-6.5$. Check: $\frac{3}{-7.5}=-0.4$ and $\frac{-4}{-10}=-0.4$ ✓; -6.5 is in the domain. Savvas quiz item 3, DOK 2, skill A87; IXL CHP. Taught by Example 1 (cut in the fast packet; TE Practice 32, the same structure, is tagged SEE EXAMPLE 1); rehearsed by the packet's Practice 3 (TE Practice 32, $\frac{3}{x+1}=\frac{2}{x-3}$, answer D, $x=11$) and Practice 2 (TE Practice 24, multiplying by both binomial denominators). **Bonus 5:** 3 for $3(2x+3)=4(x-1)$ (multiplying by both denominators or cross-multiplying), 2 for $x=-6.5$.

4. Solution: $x=5$**; extraneous solution:** $x=-3$**.** Domain: $x\neq -3$. Multiply both sides by $5(x+3)$: $-15=5x-x(x+3)=5x-x^2-3x$, so $x^2-2x-15=0$, $(x-5)(x+3)=0$, $x=5$ or $x=-3$. -3 makes the denominator $x+3$ equal to 0, so it is not in the domain of the original equation. Check $x=5$: $\frac{-3}{8}$ and $\frac{5}{8}-1=-\frac{3}{8}$ ✓. Savvas quiz item 4, DOK 2, skill A87; IXL CHP and CJC. Taught by Example 4 (kept; Part B, $\frac{3}{x-3}=\frac{x}{x-3}-\frac{x}{4}$, has the same structure). **10 points:** 3 for clearing the fractions with $5(x+3)$, 3 for $x^2-2x-15=0$ factored, 2 for $x=5$, 2 for naming -3 as extraneous with the reason (denominator 0 / not in the domain). Listing both 5 and -3 as solutions earns at most 6.

Items left off this form (available for a retake or Form B): Savvas item 1, solve $\frac{1}{2x-3}=4$ (choices A $\frac{1}{2}$, B $\frac{8}{13}$, C $\frac{11}{8}$, D $\frac{13}{8}$; answer D, $\frac{13}{8}$; DOK 2, A87; Example 1, cut in the fast packet; its one-denominator move is the first step of Example 4A); Savvas item 5, the boat in a stream: speed 10 mi/h in still water, 5 miles upstream and 5 miles back in 1 hour 20 minutes (answer D, 5 mi/h, from $\frac{5}{10-c}+\frac{5}{10+c}=\frac{4}{3}$, $c^2=25$, the negative root rejected; DOK 2, A87; Example 5, cut in the fast packet).

IXL replacement: SmartScore 70 or above on CHP and CJC replaces this quiz with 20/20.